

IQALUIT, NU, Canada

WMO: 719090

Lat: 63.7478N

Lon: 68.5444W

Elev: 34

StdP: 100.92

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-37.0	-35.6	-41.7	0.1	-36.9	-40.2	0.1	-35.5	18.1	-17.9	16.0	-21.3	3.1	320	1.007

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	17.1	11.5	14.3	10.2	12.3	9.1	12.1	16.0	10.6	14.0	9.3	11.9	6.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	7.5	13.6	8.4	6.9	12.0	7.3	6.4	10.5	34.3	16.1	30.8	14.1	27.8	12.0	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	12.7	11.2	DB	-38.9	21.3	2.4	2.4	-40.7	23.0	-42.1	24.4	-43.4	25.7	-45.2	27.4	(4)
(5)			WB	-39.0	14.2	2.4	1.6	-40.8	15.3	-42.2	16.3	-43.5	17.2	-45.3	18.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.1	-25.4	-26.1	-21.8	-12.6	-2.8	3.8	8.1	7.4	3.0	-2.9	-10.4	-18.3	(6)
(7)		DBStd	13.25	7.05	6.85	7.03	6.34	4.08	2.66	3.24	2.48	2.43	3.92	6.25	7.89	(7)
(8)		HDD10.0	6626	1097	1012	987	679	398	186	78	88	210	401	613	877	(8)
(9)		HDD18.3	9638	1355	1245	1246	929	656	435	317	338	460	659	863	1136	(9)
(10)		CDD10.0	28	0	0	0	0	0	1	19	8	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	4.7	4.4	4.5	4.8	4.8	4.2	3.6	3.8	4.4	5.0	5.3	4.8	(14)
(15)	Precipitation	PrecAvg	375	17	16	25	23	24	45	46	60	47	28	18	18	(15)
(16)		PrecMax	536	66	43	116	81	62	142	131	105	105	69	53	36	(16)
(17)		PrecMin	150	0	3	3	0	5	3	10	17	11	1	1	1	(17)
(18)		PrecStd	78	14	12	22	18	13	29	26	27	24	16	11	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	-2.5	-0.6	2.5	6.3	16.4	22.5	18.8	11.7	5.9	2.2	1.7	(19)
(20)			MCWB	-0.3	-3.5	-1.6	1.0	3.7	10.7	14.4	12.1	7.7	4.0	1.1	0.8	(20)
(21)		2%	DB	-6.3	-8.1	-5.0	0.3	4.1	12.0	18.9	15.6	9.2	4.0	0.5	-0.7	(21)
(22)			MCWB	-7.2	-8.9	-5.7	-0.5	2.1	8.0	12.8	11.0	6.2	2.6	-0.3	-1.4	(22)
(23)		5%	DB	-10.3	-12.5	-8.3	-1.5	2.9	9.6	15.8	13.2	7.7	2.8	-0.8	-3.2	(23)
(24)			MCWB	-11.0	-13.0	-8.9	-2.4	1.3	6.5	11.2	9.5	5.5	1.6	-1.6	-4.1	(24)
(25)		10%	DB	-14.4	-16.0	-11.1	-3.1	2.0	7.9	13.4	11.6	6.5	1.8	-2.0	-6.5	(25)
(26)			MCWB	-14.9	-16.5	-11.7	-3.9	0.8	5.5	9.9	8.8	4.7	0.7	-2.8	-7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.4	-3.2	-1.6	1.3	3.9	11.0	15.0	13.0	8.1	4.3	1.2	0.9	(27)
(28)			MCDWB	0.2	-2.5	-0.7	2.2	5.8	15.9	21.1	17.0	11.0	5.5	1.8	1.8	(28)
(29)		2%	WB	-7.2	-8.8	-5.7	-0.5	2.4	8.3	13.1	11.3	6.7	2.8	0.0	-1.4	(29)
(30)			MCDWB	-6.3	-8.1	-5.1	0.2	3.6	11.5	18.2	14.9	8.7	3.7	0.4	-0.8	(30)
(31)		5%	WB	-11.0	-13.0	-8.9	-2.3	1.6	6.8	11.5	10.0	5.7	1.8	-1.5	-4.1	(31)
(32)			MCDWB	-10.4	-12.6	-8.4	-1.7	2.5	9.3	15.1	12.7	7.3	2.5	-0.8	-3.3	(32)
(33)		10%	WB	-14.9	-16.5	-11.7	-3.9	0.9	5.6	10.1	8.9	4.8	1.0	-2.8	-7.2	(33)
(34)			MCDWB	-14.4	-16.0	-11.2	-3.2	1.8	7.8	13.3	11.1	6.3	1.7	-2.0	-6.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	8.1	7.8	4.8	5.3	6.8	6.1	4.5	4.5	6.2	7.2	(35)
(36)			MCDBR	9.6	9.5	9.7	6.7	5.1	8.9	11.2	9.3	6.7	4.2	5.1	6.3	(36)
(37)			MCWBR	9.2	9.1	9.3	6.0	4.1	5.7	6.4	5.7	4.6	3.5	5.0	6.1	(37)
(38)		5% WB	MCDBR	9.6	9.4	9.5	6.6	4.7	8.5	10.7	8.7	6.1	3.8	5.1	6.3	(38)
(39)			MCWBR	9.1	9.1	9.2	6.1	3.9	5.7	6.5	5.7	4.4	3.4	5.2	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.171	0.230	0.266	0.299	0.311	0.325	0.328	0.322	0.290	0.270	0.184	0.161	(40)	
(41)		taud	1.953	2.183	2.199	2.154	2.224	2.416	2.508	2.538	2.509	2.238	1.966	1.899	(41)	
(42)		Ebn at Noon	460	723	834	877	898	894	883	858	819	645	411	218	(42)	
(43)		Edh at Noon	28	57	87	116	119	101	90	80	66	54	26	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.19	0.95	2.70	4.96	6.19	5.38	4.93	3.57	1.98	0.79	0.24	0.08	(44)	
(45)		RadStd	0.01	0.06	0.11	0.20	0.39	0.65	0.53	0.42	0.22	0.12	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+1.69	+2.02	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page